

Using Learning with Rounding to Instantiate Post-Quantum Cryptographic Algorithms

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Abstract. The Learning with Rounding (LWR) problem, introduced as a deterministic variant of Learning with Errors (LWE), has become a promising foundation for post-quantum cryptography. This Systematization of Knowledge (SoK) paper presents a comprehensive survey of the theoretical foundations, algorithmic developments, and practical implementations of LWR-based cryptographic schemes. We introduce LWR within the broader landscape of lattice-based cryptography and post-quantum security, highlighting its advantages such as reduced randomness, improved efficiency, and enhanced side-channel resistance. We explore the evolution of security reductions from LWR to LWE, including recent advances that support practical parameter regimes and address challenges in both bounded and unbounded sample settings. This paper systematically reviews existing LWR-based schemes — including Saber, Lizard, Florete, Espada, Sable, and SMAUG — analyzing their design choices, parameter sets, and performance trade-offs. Furthermore, we examine the impact of LWR on side-channel resistance, failure probabilities, and masking efficiency, demonstrating its suitability for secure and efficient implementations. By consolidating the research spanning theory and practice, this SoK aims to guide future cryptographic design and standardization efforts leveraging LWR.

Keywords: Post-Quantum Cryptography · Learning with Rounding · Lattice-Based Cryptography · Cryptographic Survey

1 Introduction

It is well-known that the recent advancements in quantum computing pose a significant threat to the classical public-key cryptographic systems, such as RSA [RSA78] and ECC [Kob87, Mil86], which rely on the hardness of problems like integer factorization and discrete logarithms. As quantum algorithms [Sho94, PZ03] can efficiently solve these

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problems, the cryptographic community has been working to develop quantum-resistant alternatives. This realm of post-quantum cryptography (PQC) aims to design cryptographic schemes secure against both classical and quantum adversaries.

In response to this challenge, the National Institute of Standards and Technology (NIST) initiated a multi-year standardization process to evaluate and select quantum-safe cryptographic algorithms. This effort culminated in the selection of several algorithms for standardization, including the recommended schemes ML-KEM [MLK24] (based on the CRYSTALS-Kyber [BDK⁺18b] key encapsulation mechanism) and ML-DSA [MLS24] (based on the CRYSTALS-Dilithium [DKL⁺18] digital signature scheme). Both of these schemes are grounded in the Learning with Errors (LWE) problem [Reg05], a lattice-based hardness assumption that has become a cornerstone of modern PQC due to its strong security guarantees and versatility.

This SoK focuses on a derandomization technique for LWE introduced in 2012 by Banerjee, Peikert and Rosen [BPR12] denoted *Learning with Rounding* (LWR). LWR offers a deterministic, noise-free alternative to LWE, replacing the addition of a random error with a rounding operation. This subtle shift yields several practical advantages, including simpler implementations, reduced randomness requirements, and improved efficiency and side-channel resistance. Initially introduced for the construction of pseudorandom function families, it has been used (among others) to construct deterministic encryption [XXZ12], an IND-CCA2 secure KEM [DKRV18] and a digital signature scheme [OTF⁺20]. Over the past decade, LWR has evolved from a theoretical notion into a practical foundation for cryptographic design. Its deterministic nature makes it particularly attractive for constrained environments and high-assurance implementations, where minimizing randomness and maximizing resistance to side-channel attacks are critical. Moreover, the ability to instantiate LWR with power-of-two moduli enables efficient implementations without the need for expensive modular reduction, further enhancing its appeal for real-world deployment.

This practical relevance is highlighted by its adoption in major standardization efforts. Notably, the LWR-based Saber key encapsulation mechanism [DKRV18] is one of the finalists in the NIST Post-Quantum Cryptography standardization process, reflecting its strong balance of security, efficiency, and implementation simplicity. More recently, SMAUG-T [HCCSS24], a hybrid scheme that uses LWR for encapsulation and LWE for key generation, has been selected as the national post-quantum cryptographic standard in South Korea. These developments demonstrate that LWR is not only of theoretical interest but also a viable and competitive foundation for real-world cryptographic deployments.

Systematization Contributions. This SoK paper makes several key contributions to the study and application of LWR in post-quantum cryptography. First, it provides a comprehensive survey of the theoretical foundations of LWR, including its security reductions to LWE, parameter selection strategies, and recent advances in both bounded and unbounded sample settings. Second, it systematically reviews a wide range of LWR-based cryptographic schemes — including (u)Saber, Lizard, Florete, Espada, Sable, and SMAUG(-T) — highlighting their design choices, performance characteristics, and trade-offs. Third, the paper offers a detailed analysis of implementation aspects, including side-channel resistance, masking techniques, and decryption failure probabilities, with Saber serving as a representative case study. Fourth, it evaluates the practical efficiency of countermeasures against side-channel attacks and discusses the implications of LWR’s deterministic structure for high-assurance implementations. A primary goal of this SoK is to consolidate the fragmented research across theory and practice, providing a valuable resource for cryptographic designers and standardization bodies aiming to build secure, efficient, and quantum-resistant systems based on LWR.

2 Preliminaries

2.1 Notation

The ring of integers modulo and integer q is denoted as $\mathbf{Z}_q = \mathbf{Z}/q\mathbf{Z}$. Similarly, the polynomial quotient ring is denoted as $\mathbf{R}_q = \mathbf{Z}_q[x]/(f(x))$ where $f(x)$ in this paper is chosen to be the cyclotomic polynomial $f(x) = x^n + 1$ for a power-of-two value of n . ζ_n denotes a primitive n -th root of unity. In algorithm blocks, bold lowercase letters, such as $\mathbf{a}$, represent vectors of polynomials in the ring, while bold uppercase letters, such as $\mathbf{A}$, represent matrices. The uniform distribution over a set S is denoted by $\mathcal{U}(S)$. The inner product between two vectors a and s is written as $\langle \mathbf{a}, \mathbf{s} \rangle$. The rounding operation is denoted as $\lfloor a \rfloor_p \in \mathbf{Z}_p$.

2.2 Learning with Errors

Understanding the theoretical and practical aspects of LWR requires an introduction to the learning with errors (LWE) problem [Reg05] and their more efficient popular variants: ring learning with errors [LPR10] and module learning with errors [BGV12, LS15]. We refer the interested reader to the comprehensive surveys related to LWE [Reg10, Pei16].

Let $\mathbf{K}$ be a number field of degree n and let $\mathbf{R} = \mathcal{O}_K$ be its ring of integers and denote $\mathbf{R}_q = \mathbf{R}/q\mathbf{R}$. Let the module rank d be a positive integer and let χ denote a distribution over $\mathbf{R}$. Let $s \in \mathbf{R}_q^d$ be a secret vector and define the distribution $A_{s,\chi}$ as

$$A_{s,\chi} = (a, b = \langle a, s \rangle + e \bmod q) \in \mathbf{R}_q^d \times \mathbf{R}_q$$

where $a \leftarrow \mathcal{U}(\mathbf{R}_q^d)$ is sampled uniformly at random and $e \leftarrow \chi$ is the error term.

The *search LWE problem* is given access to polynomially many independent samples from $A_{s,\chi}$, recover the secret s , while the *decision LWE problem* is to distinguish between samples drawn from $A_{s,\chi}$ and samples drawn uniformly from $\mathbf{R}_q^d \times \mathbf{R}_q$. In practice, one often takes $\mathbf{R}_q = \mathbf{Z}_q[x]/(f(x))$ with $f(x)$ the cyclotomic polynomial

$$f(x) = \Phi_{\hat{n}}(x) = x^n + 1$$

with $\hat{n} = 2n$ and n a power of two. In this Ring-LWE variant all operations are performed within the ring $\mathbf{R}_q$. A single Ring-LWE sample embeds n Plain-LWE samples worth of information while drastically reducing the size of cryptographic keys. Furthermore, the choice of the cyclotomic polynomial allows for the polynomial multiplication to be computed very efficiently using the Number Theoretic Transform (NTT) [Pol71], a variant of the Fourier Transform that involves only integer arithmetics.

The Module-LWE problem [BGV12, LS15] generalizes both Plain-LWE and Ring-LWE, offering a flexible intermediate structure which enables a trade-off between security and efficiency. In Module-LWE, the secrets are vectors of polynomials and a Module-LWE sample consists of $(\mathbf{a}, b)$, where $b = \langle \mathbf{a}, \mathbf{s} \rangle + e$. Hence, Module-LWE is a generalization of the Plain- and Ring-LWE versions. Setting the rank $d = 1$ particularizes to the Ring-LWE problem whereas setting the ring to be $\mathbf{Z}$ and the rank $d = n$ particularized to Plain-LWE problem. Additionally, for a given parameter choice of a Module-LWE, the difficulty of the problem, i.e., the security of the scheme, can be adjusted by simply tuning the module rank.

The practical values of the key ingredients from the LWE variants are listed in the table below.

	Plain-LWE	Ring-LWE (2^k -cyclotomics)	Module-LWE
Number field K	$\mathbf{Q}$	$\mathbf{Q}(\zeta_{\hat{n}})$	General number field K
Ring $\mathbf{R}$	$\mathbf{Z}$	$\mathbf{Z}[\zeta_{\hat{n}}] = \mathbf{Z}[x]/(\Phi_{\hat{n}}(x))$	$\mathcal{O}_K$ (ring of integers)
Module rank d	n	1	$d > 1$
Secret $s \in$	$\mathbf{Z}_q^n$	$\mathbf{R}_q$	$\mathbf{R}_q^d$
Public $a \in$	$\mathbf{Z}_q^n$	$\mathbf{R}_q$	$\mathbf{R}_q^d$

It should be noted that more algebraically structured LWE variants have been introduced and studied. This includes polynomial-LWE [SSTX09, RSW18], order-LWE [BBPS19], and middle-product LWE [RSSS17]. A general framework of all these LWE variants over commutative base rings is given in [PP24].

In practice, the most well-known (module-)LWE based post-quantum cryptographic algorithms are CRYSTALS-Kyber [BDK⁺18b] (which became the ML-KEM NIST standard [MLK24]) and CRYSTALS-Dilithium [DKL⁺18] (which became the ML-DSA NIST standard [MLS24]): the recommended post-quantum secure key-encapsulation mechanism and digital signature algorithms, respectively.

3 LWR Security and Parameter Selection

Introduced in [BPR12], learning with rounding can be viewed as a derandomization technique for LWE. Recall from Section 2.2 that in LWE one computes $\langle a, s \rangle + e \bmod q$: the error term e is *added* to the inner product (modulo q). Instead of this addition, the idea of LWR is to *scale and round* this inner product to some smaller group $\mathbf{Z}_p$. Specifically, this rounding divides the elements in $\mathbf{Z}_q$ up into p contiguous intervals and maps an element of $\mathbf{Z}_q$ to its corresponding index in $\mathbf{Z}_p$. The value of p needs to be smaller than q otherwise no error is introduced. This rounded inner product $\left\lfloor \frac{p}{q} \langle a, s \rangle \right\rfloor$ is denoted as

$$\lfloor \langle a, s \rangle \rfloor_p = \left\lfloor \frac{p}{q} \langle a, s \rangle \right\rfloor \in \mathbf{Z}_p.$$

Here $\mathbf{Z}_q$ (resp. $\mathbf{Z}_p$) is represented by $[0, q[$ (resp. $[0, p[$). Note that when $p \mid q$, the rounded version of the uniform distribution on $\mathbf{Z}_q$ is again uniform on $\mathbf{Z}_p$. The rounding operation can be extended to vectors / polynomials by applying it coefficient wise.

Analogously to the ring- and module-variants of learning with errors, one can define ring- and module-versions of LWR. A table showing the differences (and similarities) between plain-, ring- and module LWR can be found below and follows the one for LWE.

	Plain-LWR	Ring-LWR (cyclotomics)	Module-LWR
Number field K	$\mathbf{Q}$	$\mathbf{Q}(\zeta_{\hat{n}})$	General number field K
Ring $\mathbf{R}$	$\mathbf{Z}$	$\mathbf{Z}[\zeta_{\hat{n}}] = \mathbf{Z}[X]/\Phi_{\hat{n}}(X)$	$\mathcal{O}_K$ (ring of integers)
Module rank d	n	1	$d > 1$
Secret $s \in$	$\mathbf{Z}_q^n$	$\mathbf{R}_q$	$\mathbf{R}_q^d$
LWR-sample $(a, b) \in$	$\mathbf{Z}_q^n \times \mathbf{Z}_p$	$\mathbf{R}_q \times \mathbf{R}_p$	$\mathbf{R}_q^d \times \mathbf{R}_p$

The decision version of the Learning with Rounding problem is as follows.

Definition 1 (Decision Learning with Rounding, based on [BPR12, Definition 3.1]). Let $n \geq 1$ and moduli $q > p \geq 2$ be integers. Let $\mathbf{R}_q = \mathbf{Z}_q[x]/f(x)$ be as defined above.

- **(Plain-LWR)**: For a vector $s \in \mathbf{Z}_q^n$, define the LWR distribution L_s to be the distribution over $\mathbf{Z}_q^n \times \mathbf{Z}_p$ obtained by choosing a vector $a \leftarrow \mathbf{Z}_q^n$ uniformly at random, and outputting $(a, b = \lfloor \langle a, s \rangle \rfloor_p)$.
- **(Ring-LWR)**: For a polynomial $s \in \mathbf{R}_q$, define the ring-LWR distribution L_s to be the distribution over $\mathbf{R}_q \times \mathbf{R}_p$ obtained by choosing $a \leftarrow \mathbf{R}_q$ uniformly at random and outputting $(a, b = \lfloor a \cdot s \rfloor_p)$.
- **(Module-LWR)**: Fix a module rank $d > 1$. For $s \in \mathbf{R}_q^d$, define the module-LWR distribution L_s to be the distribution over $\mathbf{R}_q^d \times \mathbf{R}_p$ obtained by choosing $a \leftarrow \mathbf{R}_q^d$ uniformly at random and outputting $(a, \lfloor \langle a, s \rangle \rfloor_p)$.

The **decision (ring/module-) LWR** $_{n,q,p}$ problem is to distinguish (with non-negligible advantage in n) between polynomially many independent samples $(a_i, b_i) \leftarrow L_s$ and the same number of samples drawn uniformly and independently from $\mathbf{Z}_q^n \times \mathbf{Z}_p$, $\mathbf{R}_q \times \mathbf{R}_p$, or $\mathbf{R}_q^d \times \mathbf{R}_p$, respectively.

The high-level idea for the security of LWR is to show that the security is statistically close to that of solving decision-LWE (distinguishing from random $\mathbf{u}$) [AKPW13]:

$$(a, \lfloor \langle a, s \rangle \rfloor_p) \stackrel{\text{stat}}{\approx} (a, \langle a, s \rangle + e) \stackrel{\text{comp}}{\approx} (a, \mathbf{u}).$$

This security of LWE was initially studied in [BPR12]. This paper shows the following result.

Theorem 1 (Reduction to LWE [BPR12, Theorem 3.2]). *Let χ_n be any efficiently sampleable family of probability distributions, indexed by the security parameter n , over $\mathbf{Z}$ which is $\beta(n)$ -bounded (that is, $\Pr_{x \leftarrow \chi}[|x| > \beta] \leq \text{negl}(n)$). Let $q \geq p \cdot \beta \cdot n^{\omega(1)}$. Then for any distribution over the secrets $s \in \mathbf{Z}_q^n$ solving decision-LWR $_{n,q,p}$ is at least as hard as solving decision-LWE $_{n,q,\chi}$ for the same distribution over s .*

This security result requires that the modulus $q \geq p \cdot \beta \cdot n^{\omega(1)}$ is super-polynomial. This result was improved by [AKPW13] by providing a security reduction that works for a larger range of parameters: the size of the modulus q used is not super-polynomial but polynomial [AKPW13, Thm. 4.1+Cor. 4.2]. Specifically, the authors show the hardness of LWR (using parameters n, m (the number of samples), q , and p) assuming the hardness of LWE parameters n', m, q, β as long as

$$n \geq \frac{\log(q)}{\log(2\gamma)} \cdot n' \quad \text{and} \quad q \geq \gamma(nm\beta p)$$

for some parameter $\gamma \geq 1$. For the reduction to work the modulus q needs to be much larger than the number of LWR samples m . Hence, when an attacker is allowed to choose the number of LWR samples (and is not fixed a priori) the modulus q becomes super-polynomial again. The modulus q needs to be prime or the above inequality should hold for the largest prime divisor $q_{\max}$ of q . Moreover, the result requires q and $q/q_{\max}$ to be co-prime: this implies that the result does not hold when the modulus q is a power of two.

This restriction is lifted by the results from [BGM⁺16]. Firstly, the bound on the modulus q is improved to

$$q \geq 2m\beta p$$

and secondly, this result holds without any number-theoretic restrictions on q : specifically, the results holds when the modulus q is a power of two [BGM⁺16, Thm. 1 + Thm. 3].

Theorem 2 (One-wayness of LWR [BGM⁺16, Theorem 1]). *Let p , q , n , m , and β be integers such that $q > 2p\beta$. For every algorithm Learn ,*

$$\Pr_{A,s,e} [\text{Learn}(A, \lfloor As + e \rfloor_p) = s] \geq \frac{\Pr_{A,s} [\text{Learn}(A, \lfloor As \rfloor_p) = s]^2}{(1 + 2p\beta/q)^m},$$

where $A \leftarrow \mathbf{Z}_q^{m \times n}$, the noise e is independent over all m coordinates, β -bounded and balanced in each coordinate, and s is chosen from any distribution supported on $\mathbf{Z}_q^{n*}$.

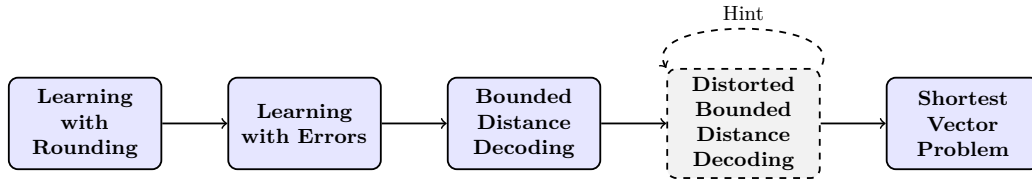
This result proves the hardness of LWR when the number of samples m are bounded. More recently, the authors of [NR23] studied the unbounded setting and in this regime identify a barrier using “pointwise” reductions from LWE to LWR. The main result of this work and the associated conjecture require the modulus q to be prime [NR23, Thm. 1 + Conj. 1]: something which in practical instantiations is not the case (cf. Section 4). It should be noted that the results of [NR23] indicate new ideas are necessary to prove the security in the unbounded sample setting and *does not* suggest that LWR is easy.

3.1 Parameter Selection in Practice

In practice, parameters for all lattice-based cryptographic schemes are chosen in such a way that the scheme is functionally correct with overwhelming probability (see Section 6 on failure probabilities), efficiency is optimized (in terms of running time as well as memory usage and key sizes) and the security goals are met (with some margin for future improvements in compute power and algorithm advancements). In general in cryptography, this is not a trivial task [LV01]. As shown in the previous section, the security of LWR can be reduced to LWE, which in turn can be related to quantumly solving the unique shortest vector problem which is believed to be a hard problem [vEB81, Mic98]. However, to select *practical* parameters one often estimates how long a *practical* attack would take and relates this to a concrete security level.

The security of lattice problems has a vast body of literature starting with [Ajt96] which linked the average case complexity of lattice problems to the worst case: i.e., a random instance of the shortest vector problem in a lattice is hard. Many algorithms for solving LWR and LWE rely on lattice reduction as the central step. Depending on the block size used, and the quality of output required, one typically estimates the running time of the lattice reduction algorithm BKZ [SE94] and improvements including [GNR10, CN11].

The most popular approach for such an assessment is using the “`lattice-estimator`” which provides functions for estimating the concrete security of LWE and LWR instances [APS15]. For LWR such a high-level chain of reductions look like



where the dashed box is optional and integrates hints into lattice reduction attacks as explained in [DDGR20]. The best known attacks are then used to solve this shortest vector problem and practical parameters are based on the running times for such algorithms.

Conclusions and Future Directions

The theoretical foundations of Learning with Rounding have matured significantly since its introduction as a derandomized variant of Learning with Errors. In this section we

have highlighted the core security reductions from LWR to LWE, the evolution of these reductions to support more practical parameter regimes, and the challenges that arise in both bounded and unbounded sample settings. The evolution of the modulus q from being super-polynomial (in terms of the security parameter) in size and prime to polynomial and allowing the usage of a powers-of-two modulus. The unbounded sample setting remains an open challenge.

Future work should aim to refine the existing reductions to yield tighter bounds and better concrete security estimates, especially for parameter sets used in practice. Moreover, the security in the unbounded sample regime should be investigated further. The limitations of pointwise reductions in this unbounded setting call for novel proof techniques or alternative hardness assumptions that can bridge this gap.

4 LWR based Key Establishment and Signatures

Mathematically hard problems such as LWR are a great source for post-quantum secure cryptographic constructions, such as key establishment and digital signatures. To define a key establishment protocol based on LWR (or any suitable hard problem), the high-level approach is to first construct a Public-Key Encryption (PKE) scheme, and then transform it into a Key Encapsulation Mechanism (KEM) scheme by using a variant of the Fujisaki-Okamoto (FO) transform [FO99, FO13]. The FO transform turns a passively (IND-CPA) secure PKE scheme into an efficient chosen-ciphertext attack (IND-CCA) secure KEM. This design approach has been followed for the two NIST PQC KEM-finalists Saber [DKRV18] and Kyber [BDK⁺18b] based on LWR and LWE, respectively. As the choice of FO transform is largely orthogonal to the selection of the PKE, we discuss these steps separately in Section 4.2. While the number of digital signature designs based on LWR is much smaller than for KEMs, we include them for completeness.

4.1 Public-Key Encryption (PKE)

There are several choices involved in the design of a public-key encryption scheme based on a KEM. Generally speaking, modern public-key encryption schemes based on LWR (and LWE) follow the approach from Regev [Reg05]. We refer to Algorithm 1 for a general description. In the remainder of this section we will introduce the various aspects of this public key encryption scheme and how they can be instantiated. We then return to the different proposals that have been defined in recent years, see Table 2 for an overview.

4.1.1 Security Reduction

A security reduction for Algorithm 1 can be constructed similar to the security reduction of Saber [DRV18]¹. This proof reduces the security of the public-key encryption scheme to the hardness of LWR, using similar secret distributions, ring structures, and moduli q and p . Algorithm 1 introduces a minor variation that allows multiple bits to be encrypted per coefficient; this modification does not impact the underlying proof strategy since it only impacts the decryption failure probability (increasing with the number of encrypted bits B). A machine-verifiable version of the Saber proof has later been provided in [HMS22].

Informally, the security reduction can be stated as follows:

Theorem 3. *In the random oracle model where $\mathbf{A}$ is assumed to be generated through a random oracle, for any adversary $\mathcal{A}$, there exist two adversaries $\mathcal{B}_0$ and $\mathcal{B}_1$ with roughly the same running time such that $\text{Adv}_{\text{Saber.PKE}}^{\text{ind-cpa}}(\mathcal{A}) \leq \text{Adv}_{\text{keygen}}^{\text{mod-lwr}}(\mathcal{B}_0) + \text{Adv}_{\text{encrypt}}^{\text{mod-lwr}}(\mathcal{B}_1)$, if $q/p \leq p/T$.*

¹The original proof contained a minor error, which has been corrected in the updated ePrint version.

Algorithm 1 Generic Public-Key Encryption (PKE) scheme based on module-LWR.

MLWR-PKE.Keygen
Input: System parameters q, p, ℓ, D_s
Output: (pk, sk)

- 1: $\mathbf{A} \leftarrow \mathbf{R}_q^{\ell \times \ell}$
- 2: $\mathbf{s} \leftarrow D_s$ $\triangleright \mathbf{s} \in \mathbf{R}_q^\ell$
- 3: $\mathbf{b} = \lfloor \frac{p}{q} \mathbf{A}^T \mathbf{s} \rfloor \in \mathbf{R}_p^\ell$
- 4: **return** $(pk = (\mathbf{b}, \mathbf{A}), sk = \mathbf{s})$

MLWR-PKE.Encrypt
Input: $pk = (\mathbf{b}, \mathbf{A}), m \in \mathbf{R}_{2^B}$
Output: c

- 1: $\mathbf{s}' \leftarrow D_s$ $\triangleright \mathbf{s}' \in \mathbf{R}_q^\ell$
- 2: $\mathbf{b}' = \lfloor \frac{p}{q} \mathbf{A} \mathbf{s}' \rfloor \in \mathbf{R}_p^\ell$
- 3: $v' = \langle \mathbf{b}, \mathbf{s}' \rangle + \frac{q}{2 \cdot p} \in \mathbf{R}_p$
- 4: $c_m = \lfloor \frac{t}{p} \cdot (v' + p/(2^B) \cdot m) \bmod p \rfloor \in \mathbf{R}_t$ $\triangleright$ Encode B bits of m per coefficient
- 5: **return** $c = (c_m, \mathbf{b}')$

MLWR-PKE.Decrypt
Input: $(\mathbf{s}, c = (c_m, \mathbf{b}'))$
Output: m'

- 1: $v = \mathbf{b}'^T \mathbf{s} \in \mathbf{R}_p$
 - 2: $m' = \lfloor \frac{2^B}{p} ((v - \frac{p}{t} \cdot c_m + (\frac{p}{2 \cdot t} - \frac{q}{2 \cdot p})) \bmod p) \rfloor \in \mathbf{R}_{2^B}$
 - 3: **return** m'
-

where $\text{Adv}_{\text{keygen}}^{\text{mod-lwr}}$ and $\text{Adv}_{\text{encrypt}}^{\text{mod-lwr}}$ are the LWR samples constructed in the key generation and encryption procedure respectively. For a more formal analysis we refer to the Saber paper.

The security proof does not impose restrictions on the choice of ring or module rank, nor does it require particularly strict conditions on the secret distribution. However, it does rely on the condition that p divides q ; otherwise, the public component $\mathbf{b}$ would not be uniformly distributed, and consequently can not be used as public input to generate an LWR sample as this requires the public value to be uniformly distributed. The proof also also requires $q/p \leq p/T$, but this condition is typically met in practice, as optimal parameter choices generally involve a small T to minimize bandwidth. For this same reason ML-KEM does not apply rounding to the public key, whereas the initial Kyber proposal did.

Another subtlety in the proof lies in the computation of c_m . Instead of applying a standard rounding operation, the constant $q/(2p)$ is added during encryption. This adjustment ensures that the rounding operation used to compute $\mathbf{b}'$ aligns with the one used to compute c_m (in the derivation of v'), and is required for the proof. The resulting deviation from standard rounding is later corrected during decryption (line 2), which compensates for the altered mean and thus reduces the noise. This has no significant effect on the efficiency of the scheme, but it does make it slightly more complicated to implement.

The above restrictions (namely, that p divides q and the use of an adapted rounding constant) are artifacts of the security proof, and there are currently no known attacks that exploit variants of the protocol that do not enforce these constraints. It remains an open research question whether and by how much the use of a prime q in combination with public key rounding, or the use of standard rounding in the computation of c_m , affects the security of the encryption scheme.

4.1.2 (Module) Learning With Rounding: Ring and Modulus Considerations

The first and most critical design aspect of a Ring/Module-LWR based encryption scheme is its underlying ring. For the ring $\mathbf{R}_q = \mathbf{Z}_q[x]/(f(x))$ this comes down to the selection of q and f . As explained in Section 4.1.1 the security reduction requires q to be divisible by p , which is typically achieved by selected q as a power of two (which is also supported by the theoretical security reduction as explained in Section 3). Hence, the selection can be made by choosing the smallest powers of two q and $p < q$ (the rounding modulus) such that the intended security level is achieved (see Section 3.1). A notable difference compared to many LWE based designs, where q can be selected to be a prime, is that the Number-Theoretic Transform (NTT) does not apply natively due to the lack of appropriate roots of unity in $\mathbf{Z}_q$. This affects the efficiency of polynomial arithmetic in $\mathbf{R}_q$, leading to generally slower implementations of LWR than LWE. However, it was first shown in [CHK⁺21] that the asymptotically faster NTTs can still be used by embedding $\mathbf{R}_q$ into a larger NTT-friendly ring with prime modulus q' . While this would increase the bit length of the modulus by at least a factor 2, it can still lead to improved arithmetic compared to Toom-Cook or Karatsuba on certain platforms. For example, [CHK⁺21] uses 6 layers of size-256 negacyclic NTTs followed by 4x4 schoolbook multiplications using

$$q' = 25166081 = 196610 \cdot 128 + 1$$

for parameter sets Saber and Firesaber resulting in approximately 60 percent faster multiplication than in $\mathbf{R}_{8192}$. Ultimately, while there is certainly a measurable difference in the efficiency of LWR and LWE, they are close enough in performance that it is not significant for many practical applications. The selection of f and optionally the module rank ℓ must be made such that the LWR problem is sufficiently hard. Typically f is chosen as $x^n + 1$ for a certain choice of n (e.g., $n = 256, 512, 1024$) in combination with ℓ to achieve varying security levels.

Having established the ring, the distribution $D_{\mathbf{A}}$ is fixed to the uniform distribution $\mathcal{U}_q$ on $\mathbf{R}_q^{\ell \times \ell}$. The final choice that remains is that of the secret distribution $D_{\mathbf{s}}$. As the security of LWR depends on the variance of $D_{\mathbf{s}}$, and less so on its precise distribution, this leaves more room for choice. The Saber proposal follows a centered binomial distribution β_μ with coefficients in $[-\mu, \mu]$, similar to the distribution used in Kyber. Other proposals include for example the uniform distribution (uSaber) or a sparse binary distribution (RLizard); we refer to Section 4.2.1 for more details. While sparse distributions can have major advantages in terms of secret key size and efficiency of computations, their security is not as well understood.

4.1.3 Pseudo-Random Generation

While ring and module LWR reduce secret, public key and ciphertext sizes, they can still be prohibitively large for distributing or storing. As such, pseudo-random generation is used to generate their values on-the-fly. For the public key matrix $\mathbf{A}$ this typically means that it is replaced by a seed $\mathbf{seed}_{\mathbf{A}}$ from which $\mathbf{A}$ is generated. The bit length of the seed should be sufficiently large, e.g., 32 or 64 bytes. The expansion from $\mathbf{seed}_{\mathbf{A}}$ to $\mathbf{A}$ consists of the bit expansion of $\mathbf{seed}_{\mathbf{A}}$, combined with a sampling routine that uses the generated bits to sample from $D_{\mathbf{A}}$. For LWR based proposals this second step is trivial: as $D_{\mathbf{A}} = \mathcal{U}_q$ for a power of two q , the sampling can be completed by simply selecting $\epsilon_q = \log(q)$ bits per coefficient. This is in contrast to primes q where rejection sampling (or equivalent) is required, leading in particular to non-constant execution times.

Similarly, secret key sizes can be reduced by storing a seed $\mathbf{seed}_{\mathbf{s}}$ instead of the expanded value $\mathbf{s}$. This trades off storage size for efficiency during decryption, and it will be dependent on the application which is favored. Since the secret key format does not require compatibility between different parties, this selection can be made on a case by

case basis. Generally the secret key generation begins by sampling a seed $\mathbf{seed}_s$ internal to the key generation algorithm from a true randomness source, after which it is expanded to the full secret $\mathbf{s}$ sampled in D_s . The sampling routine strongly depends on the choice of D_s , which affects the efficiency of the implementation. In particular, as the secret key must potentially be protected against timing and other side-channel attacks, complexity in the sampling routines can impact the effectiveness of masking techniques.

The choice of symmetric primitive is orthogonal to the selection of the sampling distributions themselves. Often SHA-3 or SHAKE are deployed due to the convenient nature of SHAKE as an eXtensible Output Function (XOF) that allows to generate an arbitrary number of bits, and due to their approval in the NIST PQC competition. However, any sufficiently robust symmetric primitive could be used for this purpose.

4.1.4 Decryption Failures

After an execution of an LWR encryption protocol as given in Algorithm 1, there is a small probability that the decrypted message m' is different from the input message m . A *decryption failure* is defined as an occurrence where $m \neq m'$. To calculate the probability of such an event happening we define the noise introduced by the various rounding operations:

$$u = \frac{q}{p} \cdot \mathbf{b} - \mathbf{A}^T \mathbf{s}; \quad u' = \frac{q}{p} \cdot \mathbf{b}' - \mathbf{A} \cdot \mathbf{s}'; \quad u'' = \frac{q}{t} \cdot \lfloor \frac{t}{q} \cdot \mathbf{b} \mathbf{s}' \rfloor - \mathbf{b} \mathbf{s}'. \quad (1)$$

We can then compute the condition for correct decryption as:

$$-\frac{q}{2 \cdot 2^B} \leq (\mathbf{u}'^T \mathbf{s} - \mathbf{u}^T \mathbf{s}' - \mathbf{u}'')_i < \frac{q}{2 \cdot 2^B} \quad (2)$$

for each coefficient i of the polynomial. As proposed by Bos et al. [BDK⁺18a], one can compute an upper bound on this decryption failure probability by computing the full distribution of the coefficients of $(\mathbf{u}'^T \mathbf{s} - \mathbf{u}^T \mathbf{s}' - \mathbf{u}'')$, and assuming independence between the coefficients, which is a reasonable assumption as long as there is no error correction as shown in [DVV19]. Note that decryption failures depend on the secret key, and as such will leak information about the secret key. This means that for encryption schemes where a secret key is reused, the decryption failure rate typically needs to be cryptographically low. We summarize the decryption failure probabilities for existing schemes in Table 2 and will cover attacks that make use of decryption failures in Section 6.

4.2 Key Encapsulation Mechanisms (KEMs)

This Fujisaki Okamoto (FO) transform [FO99, FO13, HHK17] turns a passively-secure PKE scheme into an efficient chosen-ciphertext attack (CCA)-secure KEM. In [HHK17] several variants are analyzed that are similar yet different. They define the transforms $\text{FO}^\perp$, $\text{FO}_m^\perp$ (the ciphertext is not included in the shared secret creation), $\text{FO}_{pk}^\perp$ (the public key is included in the derivation of random coins for multi-user security) and $\text{FO}_{pk,m}^\perp$, all of which *implicitly* reject a ciphertext when re-encryption fails by returning a key that is indistinguishable from random. For each variant there is an option to *explicitly* reject by aborting for all rejected ciphertexts. While early proposals in the NIST competition selected $\text{FO}^\perp$ or $\text{FO}_{pk}^\perp$ or their explicit variants, subsequent work from [BHH⁺19] showed that the inclusion of the ciphertext (or its digest) into the key derivation is not necessary.

In the more recent work [DHK⁺21] the authors look into yet another variant $\text{FO}_{\text{ID}(\text{pk}),m}^\perp$ that includes the use of prefix hashing. The core idea behind the prefix hashing is to feed a small unpredictable part of the public-key into the hash function instead of the entire large public-key itself. This significantly speeds up the computation of the KEM where the cost of this hash computation often dominates (in software implementations). An overview

Table 1: Security reduction of $\text{Adv}_{\text{KEM}}^{(n,q_C)\text{-IND-CCA}}$ to $\text{Adv}_{\text{PKE}}^{(n,q_C)\text{-IND-CPA}}$ (denoted as Adv in the Table) in the ROM and QROM models (simplified) for the different Fujisaki-Okamoto variants. Here, $\delta(n)$ is the n -user correctness error of PKE, q_C is the number of challenges, q_F is the number of (Q)ROM queries, and e is the min-entropy of $\text{ID}(\text{pk})$. The majority of the table is taken from [DHK⁺21, Figure 1] with the addition of $\text{FO}_m^\times$ and $\text{FO}_{\text{pk}}^\times$ following the result of [BHH⁺19, Theorem 5].

FO Variant	ROM	QROM
$\text{FO}_{\text{ID}(\text{pk}),m}^\times$ [DHK ⁺ 21]	$\text{Adv} + q_F \delta(n) + \frac{n^2}{2^e}$	$\sqrt{q_F \text{Adv}} + q_F^2 \delta(n) + \frac{n^2}{2^e}$
$\text{FO}_{\text{pk},m}^\times$ [HHK17]	$\text{Adv} + q_F \delta(n)$	$\sqrt{q_F \text{Adv}} + q_F^2 \delta(n)$
$\text{FO}_{\text{pk}}^\times$ [HHK17, BHH ⁺ 19]	$\text{Adv} + q_F \delta(n)$	$\sqrt{q_F \text{Adv}} + q_F^2 \delta(n)$
$\text{FO}_m^\times$ [HHK17]	$\text{Adv} + n \cdot q_F \delta(1)$	$n q_C \cdot (\sqrt{q_F \text{Adv}} + q_F^2 \delta(1))$
$\text{FO}^\times$ [HHK17, BHH ⁺ 19]	$\text{Adv} + n \cdot q_F \delta(1)$	$n q_C \cdot (\sqrt{q_F \text{Adv}} + q_F^2 \delta(1))$

of the security advantages of such approaches is given in Table 4.2 (from [DHK⁺21]). We refer to Table 2 for an overview of the use of FO per LWR scheme.

4.2.1 Plain-, Ring-, Module-LWR Instantiations

Finally we describe various instantiations of LWR based KEMs that have appeared over the years. We focus on Ring and Module based KEMs due to their favorable key and ciphertext sizes. We highlight the unique design features and their advantages and disadvantages compared to alternatives. We summarize a high-level comparison in Table 2.

All schemes have reported performance results on popular x86 computer architectures: including reference and optimized implementations often using the single instruction, multiple data programming paradigm. Unfortunately, these benchmarks are all done on a variety of different platforms (with platform specific optimizations) making direct comparisons difficult. Furthermore, the majority of the LWR-based schemes have not been submitted to the ECRYPT benchmarking of cryptographic systems (eBACS) [sup25] for benchmark and comparison. An exception is when targeting embedded devices where the popular pqm4 [KRSS19, KKP24] project is used: a reference framework for benchmarking implementations of PQC KEM and signature schemes for Arm Cortex-M4 based embedded systems. This results in benchmark results (when available) on the same platform measured in a similar fashion. This allows for an apples-to-apples comparison. The available performance results, key and ciphertext sizes and are reported in Table 3.

(Ring) Lizard. Lizard [CKLS18] uses a mixed LWE and LWR approach. The public key is a LWE sample (with error terms sampled from a discrete Gaussian distribution) and the encryption procedure uses deterministic error: learning with rounding. Note that Lizard uses the plain version of LWE and LWR, but the proposals also include variants called Ring-Lizard (RLizard) that rely on the ring version of LWE and LWR, leading to smaller public keys and ciphertexts. A distinguishing feature of Lizard is its use of sparse binary secrets sampled from the distribution $\mathcal{H} = \mathcal{H}WT_m(h)$ sampling m elements from $\{-1, 0, 1\}$ of which exactly $m - h$ are non-zero.

There exist various instances of Lizard and RLizard. In 2016 a proposal was submitted to the NIST PQC competition [CPL⁺17] containing parameter sets for both Lizard and RLizard target the different NIST security levels. A subsequent paper [CKLS18] contains a subset of the NIST parameter sets, in particular a version of RLizard targeting the NIST Level-I parameters. This selects $n = 1024$, $\epsilon_q = 10$, $\epsilon_p = 8$ and $\epsilon_t = 8$ (not explicitly stated,

Table 2: Comparison of parameters for ring / module LWR-based post-quantum cryptographic schemes targeting 128-bit security. Using the MLWR parameters with ring $\mathbf{R} = \mathbf{Z}[x]/(f(x))$, matrix distribution $D_{\mathbf{A}}$, (ephemeral) secret key distribution $D_{\mathbf{s}}$ for $\mathbf{s}$ and/or $\mathbf{s}'$, and B bits per input message. with ϵ_q the log of q , ϵ_p the log of p , ϵ_t the log of t , ℓ the module rank (with $\ell = 1$ for ring-based schemes), and FPR the log of the failure probability. SVP classical (C) and quantum (Q) bit security are included.

Scheme	$f(x)$	ℓ	ϵ_q	ϵ_p	ϵ_t	$D_{\mathbf{A}}$	$D_{\mathbf{s}}$	B	FPR	C	Q	FO
RLiz.-104*	$x^{512} + 1$	1	10	8	5	$\mathcal{U}_q$	$\mathcal{H}$	1	-167	-	104	FO $_{\text{pk}}^{\mathcal{L}}$
RLiz.-128*	$x^{1024} + 1$	1	10	8	5	$\mathcal{U}_q$	$\mathcal{H}$	1	-145	-	128	FO $_{\text{pk}}^{\mathcal{L}}$
Saber	$x^{256} + 1$	3	13	10	4	$\mathcal{U}_q$	β_4	1	-136	189	172	FO $_{\text{pk}}^{\mathcal{L}}$
uSaber	$x^{256} + 1$	3	12	10	4	$\mathcal{U}_q$	$\mathcal{U}_2$	1	-167	182	165	FO $_{\text{pk}}^{\mathcal{L}}$
Florete	$x^{768} - x^{384} + 1$	1	10	9	3	$\mathcal{U}_q$	β_4	1	-131	-	157	FO $_{\text{pk}}^{\mathcal{L}}$
Espada	$x^{64} + 1$	12	15	13	3	$\mathcal{U}_q$	β_4	4	-167	-	128	FO $_{\text{pk}}^{\mathcal{L}}$
Sable	$x^{256} + 1$	3	11	9	4	$\mathcal{U}_q$	β_1	1	-143	-	169	FO $_{\text{pk}}^{\mathcal{L}}$
SMAUG-T*	$x^{256} + 1$	3	11	9	4	$\mathcal{U}_q$	$_{\text{spCBD}}$	1	-179	180	159	FO $_{\text{m}}^{\mathcal{L}}$

* Uses LWE for key generation and LWR for Encaps/Decaps.

Table 3: Comparison of ring / module LWR-based post-quantum cryptographic schemes targeting 128-bit security. Sizes for the public-key (PK), secret key (SK) and ciphertext (CT) are in bytes. Performance results (perf) and stack usage (stack) for KeyGen, Encapsulation and Decapsulation are reported in thousands of cycles and KiB, respectively. These figures are measured on a Cortex-M4 using the `pqm4` benchmarking framework [KPR⁺].

Scheme	PK	SK	CT	KeyGen		Encaps		Decaps	
				perf	stack	perf	stack	perf	stack
Saber	992	2304	1088	645	6.5	821	7.1	774	7.2
uSaber	992	2208	1088	-	-	-	-	-	-
Florete	896	1152	1248	439	17.8	814	18.0	953	18.0
Espada	1280	1728	1304	2343	2.8	2568	2.1	2497	2.0
Sable	896	1152	1024	745	6.0	1004	5.9	1028	5.4
SMAUG-T	1088	1312	992	-	-	-	-	-	-

but inferred from the protocol description) leading to a failure probability of 2^{-154} and is referred to as **RLizard128**. A follow-up paper [CKLS18] proposes further improvements in the implementation and introduces a variant **RLizard104** targeting 104-bit quantum security. A notable change is the introduction of $\epsilon_t = 5$ (denoted p' in the paper) in the protocol description to also include rounding on the ciphertext.

Saber. The Saber scheme [DKRV18] is the first KEM that is fully based on (module) LWR. It was submitted to the NIST PQC competition in 2016 and made it to the third round [DKR⁺20]. By fully relying on LWR no sampling from an error distribution is needed, as opposed to RLizard that still relies on LWE for key generation. The Saber design aims for simplicity and efficiency. The choice of ring $\mathbf{Z}_q[x]/(x^{256} + 1)$ with $q = 2^{13}$ is fixed across all parameter sets, varying instead the module dimension, ciphertext modulus and secret key distribution, and the arithmetic modulo a power of two q allows for straightforward implementation without modular reduction or rejection sampling. The NIST proposal

defines three parameter sets LightSaber, Saber and FireSaber corresponding to NIST security level 1, 3 and 5 respectively. Secret keys are selected from a binomial distribution β_η with $\eta = 3, 4, 5$. A drawback of the choice of q is that NTTs are not natively supported in $\mathbf{Z}_q$ and the implementations therefore initially relied on Toom-Cook. However, by embedding into a larger ring with appropriately selected modulus (about twice the bit length of q) the polynomial multiplication can still be implemented with NTTs, albeit not quite as efficient as for example in Kyber. We refer to Section 4.1.2 for more details.

uSaber. Uniform Saber (uSaber) [DKR⁺20, Section A.2] is a variant of Saber proposed in the Round 3 submission of Saber that is not part of the main submission. The main difference is that the secret vectors $\mathbf{s}$ and $\mathbf{s}'$ are sampled from the uniform distribution $\mathcal{U}_2$ instead of the binomial distribution β_μ . This means that the secret coefficients are in the interval $[-2, 1]$ instead of the interval $[-\mu, \mu]$ in Saber and can be represented with exactly 2 bits each.

The main advantage of uSaber is its simplified secret generation process, which is used during key generation, encryption, and consequently in encapsulation and decapsulation for a KEM. A key benefit is the significant reduction in the amount of (pseudo-)random bits needed to achieve a comparable secret variance, and thus a similar security level. For instance, uSaber parameter sets require only two random bits per secret coefficient, compared to the six to ten bits required by standard Saber when using binomial sampling. This leads to faster hashing, which in turn reduces both computational overhead and the attack surface. Since hashing is a highly non-linear process, this reduction is especially valuable for masked implementations.

Furthermore, uSaber eliminates the need to compute a binomial distribution. This simplification makes implementations more straightforward, particularly in masked settings where binomial sampling, being non-linear, adds considerable complexity. By removing it, uSaber not only enhances efficiency but also lowers the attack surface, resulting in a more secure and practical design.

Scabbard Suite: Florete, Espada and Sable. Building upon the improvements of LWR-based schemes in many aspects such as multiplication routines, cryptanalysis, parameter selection, etc., Scabbard [MKKV21, KNK⁺25] was proposed during the course of NIST standardization effort as a suite of three KEMs. Each of the three KEMs is tailored to improve the state-of-the-art in different aspects such as speed, area requirements, memory footprint, and parallel implementation.

The first scheme “Florete” is a Ring-LWR scheme. As opposed to Saber, which uses module rank $\ell = 3$ and polynomial degree $n = 256$ over $\mathbf{Z}_q$ for $q = 2^{13}$, Florete relies on the ideal lattice defined by the ring $\mathbf{Z}_q[x]/(x^{768} - x^{384} + 1)$ for $q = 2^{10}$ and fixes the module rank to be $\ell = 1$. Aiming for the 192-bit security level of Saber, this reduces the matrix dimension of $\mathbf{A}$ from 3×3 to a single ring element which removes the need for matrix-vector products. An advantage of this approach is that fewer pseudo-random bytes are required for the generation of $\mathbf{A}$, which can be a bottleneck of performance (particularly in software). The design is done in such a way that the multiplication routines from Saber can be fully re-used. A downside of this design is that the security compared to Saber is reduced by approximately 15 bits [MKKV21, Table 2]. A similar approach is followed to match the security levels of LightSaber and FireSaber using rings with polynomial degree 512 and 1024 respectively, as proposed in a later work [KNK⁺25].

The second scheme is called “Espada” and aims to improve efficiency by allowing for more parallelization in the algorithm. The main idea is to reduce the larger polynomial multiplication of 256 coefficients into multiple smaller multiplications consisting of 64 coefficients each. Although a 256×256 polynomial multiplication can be reduced to 64×64 polynomial multiplications through Toom-Cook or Karatsuba, the use of such algorithms does not enable the same level of parallelization and also increases the size of the multiplier due to the evaluation and interpolation stages of Toom-Cook and Karatsuba. By making

$n = 64$ the native polynomial degree of the underlying lattice, the implementation cost of the base multiplication routine is reduced, particularly in hardware [MTK⁺20] and microcontrollers. It also improves the granularity of the parameter selection. As a result, the module rank ℓ is increased from 3 to 12 to achieve a 128-bit security level (and similarly for other security levels). While this increases the total number of 64×64 polynomial multiplications that are required compared to Saber, their independence makes parallel implementation more straightforward. The reduction of n from 256 to 64 requires the selection of $B = 4$ bits per coefficient, affecting the failure probability and ultimately requiring to increase the size of q to 15 bits.

The final scheme “Sable” follows the design of Saber most closely, but changes the secret key distribution. Whereas LightSaber, Saber and FireSaber change their choice of η to select secret key coefficients in the range from the centered binomial distribution β_μ in $[-\eta, \eta]$ for $\eta = 5, 4, 3$ respectively, Sable fixes $\eta = 1$ across all parameter sets, which makes switching between different schemes more seamless. In this case, only 2 bits are required to store a secret key coefficient, making the storage requirement equal to uSaber and helping with masked implementations. Overall, this change reduces the complexity of the binomial multiplier and makes the multiplier design uniform across parameter sets if it is implemented using the maximum values of p and q . To make up for the loss in security the values of p and q are decreased accordingly. The paper remarks that the selection of p and q for Saber was using erroneous estimations as it was considering a *continuous* as opposed to the *discrete* uniform distribution in the rounding error. The LightSable, Sable and FireSable parameter sets all assume the discrete uniform distribution which enables more accurate parameter selection and security estimations.

SMAUG and Friends. SMAUG-T [HCCSS24], a merger of SMAUG [CCHY24] and TiGER [PJP⁺22], is the most recent LWR based proposal and has been selected as a winner in the Korean PQC standardization effort. Similar to Lizard and RLizard, its key generation routine reduces to LWE while the Encaps/Decaps rely on LWR. For the generation of the ephemeral encapsulation secret they use a sparse centered binomial distribution, more precisely providing a ternary value which is zero with probability $1 - 2r$, -1 with probability r and 1 with probability r for $0 \leq r < 1/2$. The value r is selected as $1/8$ or $3/16$, where the power of two in the denominator is specifically chosen to ease implementation. It should be noted that the design has been updated in between v2.0 to v4.0 as a result of non-constant-time implementation of some of its samplers, so careful implementation is required. The advantage of the sparse secrets is that they reduce decryption failures and improve the efficiency of the KEM. In v3.0 the new “TiMER” parameter set was added to the specification, aiming to further lower the decryption failure probability due to the D2 error reconciliation mechanism. By encoding each message bit into two polynomial coefficients the probability of failure reduces at the cost of additional encoding/decoding.

4.3 Digital Signatures

The work on digital signature schemes based on LWR is more limited, and there are essentially two ideas. A paper from Migliore et al. [MGTF19] discusses efficient masking techniques for Dilithium, in particular suggesting that replacing the prime modulus by a power of two would be beneficial for implementations. While not replacing LWE by LWR, this shares many of the advantages of LWR based approaches. It is shown that the arithmetic simplifies, due to the trivial reduction modulo a power of two and the removal of rejection sampling modulo a prime. Additionally, the complex decomposition base γ_1 and γ_2 are replaced by what is essentially a truncation of bits.

A second paper by Okada et al. [OTF⁺20] presents a more significant redesign of Dilithium to replace its LWE assumption by LWR. Whereas the public key for Dilithium

is generated as an LWE sample $\mathbf{t} = \mathbf{A}\mathbf{s}_1 + \mathbf{s}_2$, the proposal is to generate $t = \lfloor \frac{p}{q} \mathbf{A}\mathbf{s}_1 \rfloor$ instead, where $q = 2^{23}$ and p are powers of two. The error term $\mathbf{s}_2$ can be recovered deterministically as $\mathbf{s}_2 = t - \frac{p}{q} \mathbf{A}\mathbf{s}_1$. Similar changes are made to the computation of $\mathbf{w} = \mathbf{A}\mathbf{y}$ and $c\mathbf{s}_2$. As $\mathbf{s}_2$ is chosen not to be stored as part of the private key, but rather to be generated on-the-fly during signing, this results in a signature scheme with smaller private key and faster key generation, but slower verification and signing than Dilithium (see [OTF⁺20, Table 3]).

Conclusions and Future Directions

LWR has proven to be a strong foundation for post-quantum key establishment and, to a lesser extent, digital signatures. Key establishment schemes, particularly those following the PKE-to-KEM paradigm via the Fujisaki-Okamoto transform, have shown strong potential for post-quantum cryptography. Many schemes have been proposed demonstrating that LWR enables efficient, compact, and side-channel-resistant constructions, especially when using power-of-two moduli.

While KEMs dominate the LWR landscape, signature schemes are still underrepresented. Future work should explore native LWR-based signature designs and further optimize existing ones. In particular, it would be interesting if an LWR based signature scheme could avoid the re-computation of $\mathbf{s}_2$ during signature generation. Additionally, more consistent benchmarking across platforms and deeper analysis of rounding strategies, secret distributions, and FO variants will help guide both theoretical and practical advancements.

5 Using LWR in High-Assurance Implementations

Since the late 1990s it is publicly known that the statistical analysis of a power trace obtained when executing a cryptographic primitive might correlate to the secret key material used during execution [KJJ99]. This area of *side-channel* attacks and associated countermeasures has since then grown into a large subfield of cryptology. Examples of more advanced attacks include (but are not limited to) higher order attacks [Mes00], correlation power analyses [BCO04] and template attacks [CRR03]. Such powerful attacks which use meta-data to deduce information about the secret key material are not always in scope. The adversary needs (for example) to be able to obtain a power trace to execute a power analysis attack. However, for many everyday applications (e.g., payment, transit, or identity solutions) such attacks are relevant and countermeasures need to be devised. For example, NIST in their submission requirements and evaluation criteria for the post-quantum cryptography standardization process emphasized this property [Nat16]:

Schemes that can be made resistant to side-channel attacks at minimal cost are more desirable than those whose performance is severely hampered by any attempt to resist side-channel attacks. We further note that optimized implementations that address side-channel attacks (e.g., constant-time implementations) are more meaningful than those which do not.

Not surprisingly, among the LWR-based schemes, Saber KEM stands out as the most studied scheme since it appeared early. Hence, for a side-channel analysis of LWR-based schemes we consider Saber as a representative example.

To organize our discussion, we divide this subsection into the following parts: First, we present a classification of SCAs relevant to Saber KEM. Then, we provide two subsubsections covering countermeasures in the CPA-security and CCA-security settings respectively. Next, we comment on SCAs targeting schemes beyond Saber. Finally, we evaluate the practical efficiency of these countermeasures.

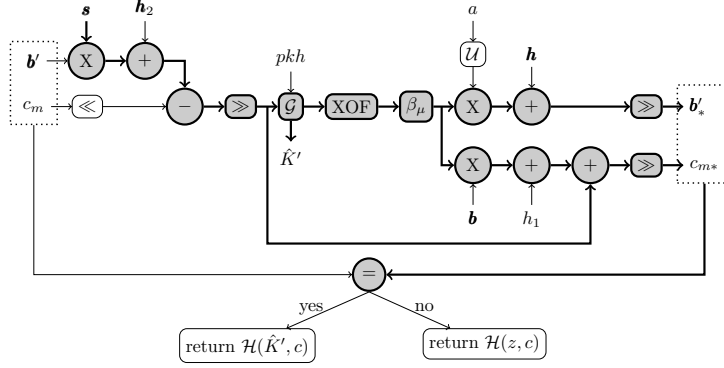


Figure 1: Decapsulation of Saber. In grey the operations that are influenced by the long term secret s and thus vulnerable to side-channel attacks [BDK⁺21].

5.1 Classification of SCA in Saber KEM

The Saber algorithm consists of two components: a CPA-secure public-key encryption scheme and a CCA-secure KEM. The key exchange protocol typically involves three stages where we suppose Alice and Bob run this protocol. First, Alice runs key generation to produce a public-secret key pair and sends the public key to Bob. Second, Bob performs the encapsulation procedure using the received public key, generating a ciphertext and a shared session key, and sends the ciphertext to Alice. Third, Alice runs the decapsulation procedure using the secret key as well as received ciphertext to recover the same shared session key. See as an example Figure 1 (from [BDK⁺21]) which displays the decapsulation of Saber and in grey all the operations which are vulnerable to side-channel attacks.

We follow the general analytical framework proposed in [RCDB24], originally developed for Kyber and Dilithium, to structure our classification of side-channel threats in the context of Saber.

Table 4: Classification of Side-Channel Attacks in Saber’s CPA-design following the framework from [RCDB24] targeting either key generation (KeyGen), Encapsulation (Encaps) or Decapsulation (Decaps) using single-trace attacks with ephemeral keys.

Attack Scenario	Target Component
Keygen	polynomial multiplication
Encaps	hashing and key derivation
Encaps	message encoding
Decaps	message decoding

5.2 Countermeasure in the Single trace SCA

In this subsection, we classify side-channel vulnerabilities in Saber’s CPA-secure design based on four critical implementation components as outlined in Table 4. Each of these components introduces specific leakage surfaces and attack opportunities for single-trace adversaries.

Polynomial multiplication. In the original Saber approach, polynomial multiplication is implemented using the Toom-Cook algorithm [Too63, Coo66]. In [LZH⁺22] it is shown that a single electromagnetic (EM) trace from Toom-Cook multiplication can be used to recover the secret key under a statistical and algebraic side-channel analysis (SASCA)

model. The attack has two phases. In the *profiling phase*, the attacker collects EM traces from a cloned device to build templates of key-dependent operations. In the *key recovery phase*, a single trace from the target is matched to these templates.

Countermeasures include applying masking to operands to break leakage correlations. Operation-level shuffling, dummy executions, and encoding randomization can add noise and uncertainty. Despite being constant-time, Toom-Cook’s regular structure remains vulnerable in high-resolution side-channel settings.

Hashing and key derivation. Like Kyber, Saber uses the Keccak-based SHAKE-128 XOF for seed expansion and key derivation, inheriting similar side-channel risks. Leakage from Keccak’s internal state can reveal input or seed data as demonstrated in [KPP20]. To mitigate such SASCA-style attacks, countermeasures like internal state shuffling, timing jitter, and targeting low-SNR platforms have been proposed to reduce trace alignment and increase noise.

Message encoding. During encryption, Saber encodes the message into polynomial coefficients, which are then compressed. The scaled coefficients differ in Hamming weight based on bit values. Early attacks on NewHope [ACLZ20] inspired a line of work targeting message encoding in lattice-based KEMs [SKL+20, XPR+22]. More recent studies focus specifically on Saber [NDGJ21, NWDP22, BNGD23], showing that these attacks remain effective under first-order and even higher-order masking, and exploit leakage from all message shares using profiling or deep learning. Thus, masking alone is insufficient: constant-time encoding, share re-randomization, or shuffling are recommended for high-security implementations.

Message decoding. During decryption, Saber reconstructs the message by scaled subtraction and rounding. Like encoding, this step manipulates each coefficient, making it vulnerable to single-trace side-channel attacks. These attacks exploit Hamming weight differences and remain effective even on masked implementations [NDGJ21, NWDP22, BNGD23, RBRC22]. Thus, shuffling or re-randomizing decoding operations is recommended for stronger protection.

5.3 Countermeasure in Multi-Trace SCA

In the multi-trace setting, the non-ephemeral key usage in Saber allows the attacker to observe multiple executions of the decapsulation procedure under the same secret. Compared to single-trace scenarios, the primary advantage is that an attacker can aggregate information across traces, either passively or through chosen ciphertexts. For a passive attacker, attacks such as correlation power analysis are accessible. In this case, the attack focuses on repeated polynomial multiplications, where intermediate values maintain consistent secret-dependent structure. As demonstrated by [MWK+24a], even sanity checks cannot fully eliminate leakage from such operations. In this case, masking remains the most effective defense.

For an active attacker, the threat surface expands further. In this case, side-channel-assisted chosen ciphertext attacks (SCA-assisted CCA) emerge, which can be classified using the four oracle models.

Binary Plaintext-Check Oracle. The attacker crafts highly sparse ciphertexts so that a specific bit of the decrypted plaintext depends exclusively on one secret-coefficient. By observing side-channel leakage during decryption, the attacker determines whether the decrypted bit equals their guess about the secret coefficient, effectively instantiating a one-bit oracle. The attacker applies this strategy repeatedly to learn the secret coefficients one-by-one. Such SCA-assisted CCA attack was introduced in [DTVV19], targeting LAC KEM via timing leakage in non-constant-time BCH decoding. It was generalized

in [RRCB20] to perform key recovery in several constant-time LWE and LWR-based KEMs by observing EM or power side-channel leakage.

In the CCA-secure KEM setting, the attacker crafts a sparse ciphertext $ct = (u, v)$ with:

$$u_i = \begin{cases} U \cdot x^0 & \text{if } i = 0, \\ 0 & \text{otherwise,} \end{cases} \quad v = V \cdot x^0,$$

where $U, V \in \mathbf{Z}^+$ are attacker-chosen. The decrypted bits m'_i then satisfy:

$$m'_i = \begin{cases} \text{Decode}(V - U \cdot s_0[0]) & \text{if } i = 0, \\ \text{Decode}(-U \cdot s_0[i]) & \text{for } i \geq 1. \end{cases}$$

Each bit of m' thus depends on a single coefficient of s_0 . If the adversary can detect whether m' matches a guess (e.g., via power leakage), a bitwise recovery oracle is formed.

Masking the decapsulation process mitigates this attack. While higher-order attacks remain possible, they require exponentially more traces.

Parallel Plaintext-Check Oracle. Parallel PC Oracle attacks extend the binary case by extracting multiple key bits per query. By crafting ciphertexts that induce secret-dependent variations in the first P bits of the decrypted message m' , the attacker reduces the number of required traces by a factor of P . The general strategy is to set the ciphertext components as:

$$u_i = \begin{cases} U \cdot x^0 & \text{if } i = 0, \\ 0 & \text{otherwise,} \end{cases} \quad v = V \cdot \sum_{i=0}^{P-1} x^i,$$

with $U, V \in \mathbf{Z}^+$. This yields $m'_i = \text{Decode}(V - U \cdot s_0[i])$, for $i \in [0, P-1]$, where each m'_i acts as a binary distinguisher for $s_0[i]$. This P -way oracle reduces trace complexity, though profiling cost grows exponentially with P due to 2^P output combinations. With access to a clone device, profiling can be done offline. Experiments show $P \leq 10$ is feasible [RRD⁺23, TUX⁺23], and full key recovery on Saber is possible with as few as 1152 traces for $P = 8$ [TUX⁺23].

As in the binary case, masking the decapsulation procedure mitigates this attack. Higher-order masking increases resistance by requiring exponentially more traces.

Decryption Failure (DF) Oracle. DF oracle-based attacks exploit the fact that decryption failures can leak information about the secret key. By crafting ciphertexts that trigger failures depending on specific secret coefficients, an attacker can recover the key through repeated queries. The first such attack, targeting FrodoKEM, used timing leakage in non-constant-time comparison logic [GJN20]. Later work extended this to power/EM side channels [BDH⁺21], showing that even masked implementations can leak distinguishable patterns during failure events. While initially demonstrated on LWE-based schemes, the same principles may apply to LWR-based KEMs like Saber, provided side-channel leakage can distinguish successful from failed decryptions. Although more traces are needed than in the binary oracle case (around 6k), the attack remains practical in unprotected, low-SNR environments.

Full Decryption (FD) Oracle. FD Oracle-based attacks represent the strongest form of chosen-ciphertext side-channel attacks, aiming to recover the entire decrypted message m' in a single query, far surpassing Binary and Parallel PC attacks. [XPR⁺22] showed that message recovery techniques exploiting leakage from encoding can be adapted to the CCA setting, enabling full key recovery on Kyber512 in as few as 6 queries. Similar attacks on Saber and other LWR-based schemes [NDJ21, WND22b, WND22a] exploit leakage from bitwise decoding, achieving full key recovery within 20 traces. In Saber, rounding during decapsulation causes bit-dependent leakage. [RBRC22] demonstrated that even with shuffling, controlled ciphertext perturbations and Hamming weight comparisons can recover

each bit. Recent work [BNGD23] further showed that profiling-based attacks can bypass combined masking and shuffling given enough queries. The only robust countermeasure is frequent key refresh, preventing attackers from accumulating sufficient traces per key instance.

Apart from the abovementioned four oracles, the polynomial multiplication where the non-ephemeral secret key is multiplied with the ciphertext can be exploited by correlation power analysis. It was first demonstrated in [MWK⁺24b]. In this work, the authors demonstrated the attack on all three lattice-based KEMs, which were the finalists of the NIST procedure. For Kyber, the multiplication happens between the secret coefficients, which are in the NTT domain. Due to this, each secret expands to a 13-bit number. However, for Saber, the secrets also expand to a larger range due to the Toom-Cook and Karatsuba transformation. But this expanded range is much smaller compared to Kyber. Due to this, a major observation in that work was that the correlation power attacks are much easier to mount on a LWR-based scheme where Karatsuba- and Toom-Cook-based multiplication is used compared to the LWE-based schemes where NTT is used.

5.4 Practical Efficiency of Countermeasures

Among countermeasures against side-channel attacks, masking is the most computationally expensive but also the most effective one against differential power analysis (DPA), especially when targeting static keys across multiple traces. Masking splits a secret s into $t+1$ random shares $(s_0, \dots, s_t)$ such that $s = \bigoplus_i s_i$ (Boolean) or $s = \sum_i s_i$ (arithmetic). Computations are performed on the shares independently, randomizing leakage and requiring $(t+1)$ th-order statistical moments for successful DPA. Early masked implementations, such as the first-order ring-LWE scheme in [RRVV15], used arithmetic masking in $\mathbf{R}_q$. While polynomial operations scale linearly with t , decoding became the main bottleneck, requiring up to 16 share refreshes per coefficient. [OSPG18] improved this with a more efficient masked decoder using distribution shifting and arithmetic to Boolean (A2B) mask conversion.

Despite these advances, first-order masking still incurs significant overhead: polynomial multiplication roughly doubles in cost, and bitwise operations like error sampling and randomness generation can be over $5\times$ slower. [BDK⁺21] observed that in masked implementations, non-linear operations (e.g., binomial sampling, hashing) dominate the cost. These often require Boolean masking, while polynomial arithmetic uses arithmetic masking: requiring mask conversions. LWR-based schemes, like Saber, benefit from two key advantages:

- Power-of-two moduli simplify mask conversions.
- Rounding is easier to mask than noise sampling.

These properties motivated the first-order masked Saber framework, achieving a $2.5\times$ overhead on ARM Cortex-M4: compared to $5\times$ for LWE-based schemes. Higher-order masked implementations of Saber and uSaber [KDvB⁺22] show that LWR schemes scale better than LWE counterparts with increasing t . uSaber is even more efficient due to its uniform secret distribution, which simplifies sampling. See Table 5.

The masked implementations of LWR-based KEMs Florete, Espada, and Sable from the suite, Scabbard, also perform better than Kyber due to the power-of-two moduli, and these schemes are portrayed as masking-friendly schemes in [KKV23]. Masked Florete and masked Sable are even faster than masked Saber for any order masking due to their efficient parameter choices. Although uSaber is shown as a mask-friendly version of Saber in [KDvB⁺22], Sable performs better than uSaber, and that also remains the same after masking, as masked Sable showed slight improvement over uSaber in performance for any order masking.

Table 5: Cycle overhead factor of a masked (t -th order) versus an unprotected implementation on Cortex-M4 [KDvB⁺22].

Masking Order t	Saber	uSaber	Kyber
1	$2.7\times$	$2.3\times$	$5.6\times$
2	$5.0\times$	$4.2\times$	$9.5\times$
3	$7.7\times$	$6.5\times$	$13.0\times$

In hardware, [FVR⁺22] proposed RISC-V instruction set extensions for masking Kyber and Saber. Their results confirm Saber’s masking efficiency over LWE schemes. [ABC⁺23] further optimized masked polynomial multiplication for power-of-two moduli. Their Saber hardware achieves first-order resistance with only $1.7\times$ area overhead and completes multiplication in just 340 cycles: significantly outperforming generic masking approaches.

Conclusions and Future Directions

LWR-based schemes, especially when using power-of-two moduli, are efficient on various software and hardware platforms. Nevertheless, these schemes remain susceptible to various side-channel attacks. Masking is one of the most effective algorithmic countermeasures against side-channel attacks, though it introduces performance costs. The deterministic structure of LWR-based schemes and the use of power-of-two moduli facilitate simpler and more efficient masking, and allows for optimized arithmetic operations. Compared to their LWE-based counterparts, LWR-based schemes generally achieve better side-channel resistance at a lower cost, making them attractive in constrained environments.

Future research on LWR-based cryptographic schemes should focus on enhancing the efficiency and scalability of masking techniques, particularly for higher-order protection. Expanding side-channel analysis beyond Saber to include other LWR-based schemes is essential for a broader understanding of their resilience. Additionally, improving automated tools for detecting side-channel vulnerabilities and verifying countermeasures, especially through formal methods, will support more robust implementations. Optimizing design choices, including secret distributions and rounding strategies, can further balance security and performance.

6 Failure Probabilities

In lattice-based encryption schemes, security is achieved by introducing an error term. The combination of these error terms, e.g. the one in the public key combined with the one from encryption, can sometimes result in protocol failures. For encryption schemes and KEMs, these failures constitute an attack vector and as such it is important to understand the potential threat of decryption failures.

6.1 Decryption failures

This section will look at decryption failures in encryption schemes, which happen when there is excessive noise due to the error terms. The main issue with these failures is that they reveal information about the secret key, as their occurrence depends on it. A failure happens if the message cannot be correctly decoded. Following the formula in Section 4.1.4, a decryption is successful if:

$$-\frac{q}{2 \cdot 2^B} \leq (\mathbf{u}'^T \mathbf{s} - \mathbf{u}^T \mathbf{s}' - \mathbf{u}'')_i < \frac{q}{2 \cdot 2^B}$$

Here, the s -terms represent the secret terms, sampled as part of the LWR problem. In contrast, the u terms are the errors introduced as part of the LWR rounding. This term is similar to the failure probability in LWE schemes, where one replaces the $\mathbf{u}$ -terms with $(\mathbf{e} + \mathbf{u})$ -terms where $\mathbf{e}$ is the error introduced due to the LWE problem.

Some of these terms are generated by Alice, while others originate from Bob. Grouping them, we can define

$$\mathcal{S} = \begin{pmatrix} -\mathbf{s} \\ \mathbf{u} \end{pmatrix}, \mathcal{C} = \begin{pmatrix} \mathbf{u}' \\ \mathbf{s}' \end{pmatrix}, \text{ and } \mathcal{G} = \mathbf{u}''.$$

In this breakdown, $\mathcal{S}$ is generated by Alice and encompasses all her secret elements within the encryption scheme. Conversely, $\mathcal{C}$ and $\mathcal{G}$ are generated by Bob. With these definitions, we can rewrite the failure equation more concisely (and using a slight approximation) as

$$\|\mathcal{S}^T \mathcal{C} + \mathcal{G}\|_\infty > \frac{q}{2^B} \quad (3)$$

As this equation demonstrates, the occurrence of a decryption failure reveals information about the secret key $\mathcal{S}$. In most scenarios, all terms within the failure equation are small, preventing modulus reduction. The above equation thus yields a linear inequality involving the secret, which can subsequently be exploited to reduce the security of the LWE or LWR sample holding the secret key.

Comparing LWE or LWR, security is defined based on the $\mathbf{e}$ and $\mathbf{u}$ terms respectively. The advantage of LWR is that the error introduced for security naturally comes with ciphertext compression. For LWE one can do additional ciphertext compression but this increases the noise with an additional $\mathbf{u}$ term and thus the failure rate.

Type of attacks. For schemes that achieve IND-CPA security but lack protection against chosen-ciphertext attacks, an adversary can readily craft ciphertexts designed to yield specific values for $\mathcal{C}$ and $\mathcal{G}$. Typically, such an adversary would generate ciphertexts with an approximate 50% probability of decryption failure. By submitting these crafted ciphertexts for decryption and observing whether a failure occurs, the adversary gains a linear equation involving the secret. By accumulating a sufficient number of these equations, the secret can be trivially recovered.

These types of chosen-ciphertext attacks were initially described in [JJ00] for NTRU, and were subsequently extended in [HNP⁺03] and [GN07]. Similar attacks targeting lattice-based homomorphic schemes have since been developed [LMSV12], and specifically on LWE-based schemes [BDH⁺19] and other works. Importantly, these techniques are adaptable for use against LWR schemes. The clear implication of these attacks is that for schemes lacking IND-CCA security, secret keys should never be reused.

The situation differs significantly for schemes offering IND-CCA security (also known as security against chosen-ciphertext attacks). The aforementioned attacks rely on specially crafted ciphertexts, and can therefore be prevented by restricting an adversary’s ability to choose ciphertexts. This is commonly achieved through a chosen-ciphertext transform (see Section 4.2). In such cases, attacks become considerably more complex, thereby permitting the safe reuse of secret keys. We will delve further into decryption failure attacks on IND-CCA schemes in the remainder of this section.

Using decryption failures. If a decryption failure is discovered, that is if a ciphertext is found with $\mathcal{C}$ and $\mathcal{G}$ that triggers such a failure, a linear inequality in the secret is obtained, as shown in Equation 3. Since the coefficients of all terms are typically small, no modular reduction occurs. This ensures the sample is useful and not merely another LWE/LWR sample. In [DGJ⁺19] it is demonstrated that these inequalities can be used to reduce the hardness of the primary sample protecting the secret key. The reduction in hardness of samples under hints, such as inequalities, was further formalized and generalized [DDGR20], which also provided an open-source codebase for calculating security after hints are applied.

One of the challenges with this approach is that the specific coefficient that caused the failure, and whether the failure was positive or negative, remain unknown. One method for determining the relative position of errors leverages the fact that two failing ciphertexts are both aligned with the secret, and thus also roughly aligned (and correlated) with each other, as shown in [DGJ⁺19]. This effect becomes more pronounced with lower failure probabilities. The authors in [DGJ⁺19] used this correlation to estimate the relative failure position for schemes with very low failure probability. Later, [DRV20] improved upon this technique by employing a belief propagation algorithm.

Interestingly, [BS20] demonstrated that while decryption failures indeed offer information about the secret key, the same holds true for successful decryptions. However, given their significantly higher probability of occurrence, the information gleaned from these successful decryptions is considerably less. Currently, all existing decryption failure attacks still depend on identifying decryption failures and do not directly process successful decryptions to extract information.

Finding decryption failures. To locate decryption failures, one can generate ciphertexts and submit them to Alice for decryption, subsequently observing her response to ascertain if a failure occurred. Due to the FO-transformation, it is crucial that ciphertexts are generated via the correct procedure, otherwise they will be rejected. A straightforward approach of generating random valid ciphertexts would result in decryption failures with a probability equivalent to the underlying scheme’s intrinsic failure probability.

In [DGJ⁺19] it is demonstrated that it is possible to generate valid ciphertexts with an increased failure probability. These are termed ‘weak ciphertexts’ and are produced by generating a large number of ciphertexts and then selecting those with high $\mathcal{C}$ and $\mathcal{G}$ norms. This method offers three key advantages: firstly, a quantum computer can accelerate this search (whereas the search for decryption failures on Alice’s machine cannot be quantumly sped up); secondly, if there is a limit on the number of queries that can be submitted, precomputation can enhance the failure rate of each query; and thirdly, if multiple targets are available, precomputation can identify weak ciphertexts applicable to multiple targets. This latter advantage is less pronounced for LWR compared to LWE. In LWE, the s' , e' , and e'' error terms are generated independently of the public key. Conversely, in LWR, only the s' term is independent of the secret key, with the error terms u' and u'' being generated based on a , which is part of the secret key.

Similar to the concept of weak ciphertexts, one can also search for weak keys. If an attacker aims to compromise the secret key of one among several parties, they can probe all parties and identify one possessing a larger-than-average $\mathcal{S}$, which consequently leads to a higher decryption failure probability. In [DGJ⁺19], an attack leveraging this property was successfully mounted while carefully limiting the number of required queries. This attack was subsequently generalised and extended in [DB22], which currently also provides the most state-of-the-art scripts for estimating the efficiency of decryption failure attacks against generic schemes.

In some encryption schemes, error correction is used to reduce the failure probability. While this does indeed reduce the failure probability and make attacks more expensive, [DVV19] showed that one has to be careful when calculating the failure probability, as the individual coefficient inputs to the error correction are correlated and thus the error correcting code is less efficient. In case of the LWE scheme LAC [LLJ⁺19], the calculation assuming independence lead to an underestimation of the failure probability of 2^{48} . Later, [GJY19] mounted an attack on the LAC scheme that used the fact that weak ciphertexts are also more effective for schemes with ECC.

For a successful decryption failure attack, multiple failing ciphertexts are required. In [DRV20], it is demonstrated that once a single failure is identified, it can significantly reduce the cost of finding subsequent failures through a procedure known as failure boosting. The expense of discovering failures in a generic setting is primarily attributed to finding

the initial one. In [BS20], it is further shown that even successful decryptions provide information that aids in more efficient discovery of decryption failures, which was later improved by [WX22]. However, in certain specific scenarios, such as the multi-target setting in [DB22], the cost of finding a second failure (in one particular target) is significantly larger than the cost of finding the first failure across all potential targets. As a result, the attack cost is generally the cost of finding the first failure; however, in the case of multi-target attacks, the attack cost is the combined cost of finding the first and the follow-up failures.

Provable security. Decryption failures are accounted for in the security proofs of KEM systems, as illustrated in Section 4.2. Specifically, the proofs include terms like $\delta \cdot q_F$ for classical attackers and $\delta \cdot q_F^2$ for quantum attackers. Here, δ represents the failure probability of the underlying encryption scheme, and q_F denotes the number of queries made to a random oracle or to the decryption oracle (e.g., hash function executions or queries to the server). Throughout the rest of this section, we will refer to these queries as operations.

In the classical scenario, roughly $q_F \approx \delta$ operations are needed to reach the proven security bound. Yet, in the quantum setting, only about $q_F \approx \sqrt{\delta}$ operations are required. This quadratic speedup is attributed to a worst-case scenario where Grover’s algorithm could potentially accelerate the search for decryption failures. However, we do not see a comparable quadratic speedup in real attacks, where the speedup using a quantum computer leans more towards the linear case than the quadratic case. The reason an attacker cannot fully utilize Grover’s algorithm is that, without precise knowledge of the secret key, they cannot accurately predict if an error will occur. This limits an attacker to searching for weak ciphertext attacks.

Bridging the gap between practical attacks and provable security remains an interesting open question for lattice-based schemes. Improvements towards tighter security proofs have been made in several papers [HHK17, SXY18, JZC⁺18, BHH⁺19, HKSU20, KSS⁺20, HHM22, HM24]. A significant complication is that attackers gain small pieces of information about the failure probability through successful queries, as noted by Nina Bindel and John Schanck [BS20]. This makes the search progressively more efficient with more queries, a factor that must be considered when developing a tighter security proof.

Conclusions and Future Directions

Decryption failures in LWR-based encryption schemes pose a subtle but serious threat, as even rare failures can leak information about secret keys. These failures can be exploited through chosen-ciphertext attacks. While current security proofs account for such failures, they often rely on conservative bounds that may not reflect practical attack scenarios.

Looking ahead, research should focus on narrowing the gap between theoretical security guarantees and real-world attack models. This includes refining failure probability estimates, improving the robustness of schemes against failure-based leakage, and developing tighter, quantum-aware security proofs. Additionally, practical tools for simulating and analyzing decryption failures will be essential for guiding secure parameter selection and implementation.

7 Conclusions and Looking Forward

This Systematization of Knowledge has demonstrated the maturation of Learning with rounding (LWR) as a foundational alternative for lattice-based cryptographic constructions. It has gone from a theoretical variant of LWE into a practical and competitive cornerstone of post-quantum cryptography.

The theoretical foundations have proven to be as mature as those of LWE as well as enable more practical parameter selection. We have surveyed the landscape of LWR constructions and analyzed the security and implementation tradeoffs. Moreover, they have shown their suitability to achieve security, efficiency and compactness of implementation, and usability of their constructions.

Currently, the landscape of LWR schemes is dominated by KEMs where particularly power-of-two moduli constructions outstand in efficiency of implementations as well as protection against side-channel attacks. Moving forward, there is a need to develop native LWR-based signature schemes to complete the post-quantum cryptographic needs.

In conclusion, LWR has firmly established its place in the post-quantum toolkit. By consolidating the last years' research, this SoK provides a clear foundation and a roadmap for the continued development and deployment of LWR-based cryptography in the post-quantum era.

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